

Spectroscopic Term Symbols

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1 Quantum Numbers

To completely describe an electron in an atom, four quantum numbers are needed: energy (n), angular momentum (ℓ), magnetic moment (m_ℓ), and spin (m_s).

The first quantum number describes the electron shell, or energy level, of an atom. The value of n ranges from 1 to the shell containing the outermost electron.

Table 1: Quantum Numbers

Number	Symbol	Possible Values
Principal Quantum Number	n	$1, 2, 3, 4, \dots$
Angular Momentum Quantum Number	ℓ	$0, 1, 2, 3, \dots, (n - 1)$
Magnetic Quantum Number	m_ℓ	$-\ell, \dots, -1, 0, +1, \dots, +\ell$
Spin Quantum Number	m_s	$+\frac{1}{2}, -\frac{1}{2}$

Table 2: Orbital Designations

n	ℓ	m_ℓ	No. of Orbitals	Orbital Name	No. of Electrons
1	0	0	1	1s	2
2	0	0	1	2s	2
	1	$-1, 0, +1$	3	2p	6
3	0	0	1	3s	2
	1	$-1, 0, +1$	3	3p	6
	2	$-2, -1, 0, +1, +2$	5	3d	10
4	0	0	1	4s	2
	1	$-1, 0, +1$	3	4p	6
	2	$-2, -1, 0, +1, +2$	5	4d	10
	3	$-3, -2, -1, 0, +1, +2, +3$	7	4f	14

2 Term Symbols

“The symbol which indicates electronic configuration & different energy states of atom is called term symbol.”

Term symbols are represented as $^{2S+1}L_J$.

HN Russell and FA Saunders introduced a system to fully represent the position of electrons in an atom. According to this method, in a multi-electron system, many different arrangements of electrons are possible. A term symbol is given for the ground state arrangement which follows all the laws (Hund's, Pauli, Aufbau etc.). Term symbols are assigned using the R-S coupling scale or L-S coupling scale.

2.1 The Russell–Saunders Coupling Scheme (R–S or L–S Coupling)

This scheme applies to smaller atoms ($Z < 30$) having less interaction between spin and orbital angular momentum.

The term symbol for the ground state configuration is:

$$^{2S+1}L_J$$

where:

- $2S + 1$ is the spin multiplicity,
- L is the orbital angular momentum,
- J is the total angular momentum.

All terms are determined by calculating:

1. **s–s coupling**, which gives total spin angular momentum (S)
2. **l–l coupling**, which gives total orbital angular momentum (L)
3. **L–S coupling**, which gives total angular momentum (J)

2.1.1 (1) s–s Coupling (S)

The value of the resulting spin angular momentum is obtained from the unpaired electrons in the outer orbit of the atom.

$$S = s_1 + s_2 + s_3 + \dots$$

Example: For P^3 (three unpaired electrons):

$$S = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

Spin multiplicity: $2S + 1 = 2\left(\frac{3}{2}\right) + 1 = 4$

Paired electron spins cancel due to opposite spin.

2.1.2 (2) l–l Coupling (L)

Total orbital angular momentum is the sum of the orbital angular momentum ℓ of each electron:

$$L = m_{\ell_1} + m_{\ell_2} + m_{\ell_3} + \dots$$

The value of L is assigned using the following letter designations:

L	0	1	2	3	4	5	6
Term	S	P	D	F	G	H	I

Example: For d^7 , with $m_\ell = +2, +1, 0, -1, -2$:

$$L = 2(+2) + 2(+1) + 1(0) + 1(-1) + 1(-2) = 4 + 2 + 0 - 1 - 2 = 3 \Rightarrow F \text{ term}$$

2.1.3 (3) L-S Coupling (J)

The total orbital angular momentum (L) couples with the total spin angular momentum (S) to give total angular momentum (J):

$$J = (L + S), (L + S - 1), \dots, |L - S|$$

Rule:

- For **less than half-filled** orbitals: use the **minimum** value of J .
- For **more than half-filled** orbitals: use the **maximum** value of J .

Example: For P^2 , $S = 1$, $L = 1$:

$$J = (1 + 1), \dots, (1 - 1) = 2, 1, 0$$

Since P^2 is less than half-filled, $J = 0$.

2.2 Worked Examples

2.2.1 Ground State Term for P^3 Configuration

Electron arrangement ($m_\ell = +1, 0, -1$, one electron each):

$$L = 1(+1) + 1(0) + 1(-1) = 0 \Rightarrow S \text{ term}$$

$$S = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}, \quad 2S + 1 = 4$$

$$J = |0 + \frac{3}{2}| \text{ to } |0 - \frac{3}{2}| = \frac{3}{2}$$

Ground state term : $^4S_{3/2}$

2.2.2 Ground State Term for d^2 Configuration

Electron arrangement (two electrons at $m_\ell = +2, +1$):

$$L = 1(+2) + 1(+1) = 3 \Rightarrow F \text{ term}$$

$$S = \frac{1}{2} + \frac{1}{2} = 1, \quad 2S + 1 = 3$$

$$J = (3 + 1) \text{ to } (3 - 1) = 4, 3, 2$$

Less than half-filled $\Rightarrow$ minimum $J = 2$.

Ground state term : 3F_2

2.2.3 Ground State Term for 7N

Configuration: $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$

$$L = 2(0) + 1(+1) + 1(0) + 1(-1) = 0 \Rightarrow S \text{ term}$$

$$S = -\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}, \quad 2S + 1 = 4$$

$$J = |0 + \frac{3}{2}| \text{ to } |0 - \frac{3}{2}| = \frac{3}{2}$$

$^4S_{3/2}$ (quartet S three-halves)

2.2.4 Ground State Term for ^{26}Fe

Configuration: $[\text{Ar}] 3d^6 4s^2$; considering $3d^6$: ($m_\ell = +2, +1, 0, -1, -2$, filled as $\uparrow\downarrow, \uparrow, \uparrow, \uparrow, \uparrow$)

$$L = 2(+2) + 1(+1) + 1(0) + 1(-1) + 1(-2) + 0 = 2 \Rightarrow D \text{ term}$$

$$S = 0 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + 0 = 2, \quad 2S + 1 = 5$$

$$J = |2 + 2| \text{ to } |2 - 2| = 4, 3, 2, 1, 0$$

More than half-filled $\Rightarrow$ maximum $J = 4$.

$$\boxed{{}^5D_4 \text{ (quintet D four)}}$$

2.2.5 Ground State Term for d^9 Configuration

$3d^9$: ($\uparrow\downarrow, \uparrow\downarrow, \uparrow\downarrow, \uparrow\downarrow, \uparrow$; $m_\ell = +2, +1, 0, -1, -2$)

$$L = 2(+2) + 2(+1) + 2(0) + 2(-1) + 1(-2) = 2 \Rightarrow D \text{ term}$$

$$S = \frac{1}{2}, \quad 2S + 1 = 2$$

$$J = (2 + \frac{1}{2}) \text{ to } (2 - \frac{1}{2}) = \frac{5}{2}, \frac{3}{2}$$

More than half-filled $\Rightarrow$ maximum $J = \frac{5}{2}$.

$$\boxed{{}^2D_{5/2}}$$

2.2.6 Finding L , S and J from a Given Term Symbol

For 3F :

$$2S + 1 = 3 \Rightarrow S = 1 \text{ (2 unpaired electrons); } L = 3 \text{ (F term)}$$

$$J = (3 + 1), (3 + 0), (3 - 1) = 4, 3, 2$$

For 1S :

$$2S + 1 = 1 \Rightarrow S = 0 \text{ (0 unpaired electrons); } L = 0 \text{ (S term)}$$

$$J = (0 + 0) = 0$$

3 Term Symbols for p^2 Configuration (Carbon)

The electronic configuration of C is $1s^2 2s^2 2p^2$. Since the $1s$ and $2s$ orbitals are fully filled, only the two $2p$ electrons contribute to the term symbols.

Three cases arise:

Figure 1: Electrons with Parallel Spins

(a) $L = +1, S = 1, 2S + 1 = 3$

(b) $L = 0, S = 1, 2S + 1 = 3$

(c) $L = -1, S = 1, 2S + 1 = 3$

L values $+1, 0, -1$ correspond to $L = 1$ (P term). All arrangements give $2S + 1 = 3$, hence: 3P

Figure 2 & 3: Electrons with Opposite Spins

- Group (a): L values $+2, +1, 0, -1, -2 \Rightarrow L = 2$ (D term), $S = 0$: 1D
- Group (b): $L = 0 \Rightarrow L = 0$ (S term), $S = 0$: 1S

Summary: The terms for p^2 are 3P , 1D , and 1S .

3.1 Energy Ordering by Hund's Rule

1. The term with the **highest spin multiplicity** ($2S + 1$) is most stable (lowest energy). 3P has $2S + 1 = 3$, so it is most stable.
2. Among terms with the same $2S + 1$, the one with **higher** L is more stable. 1D ($L = 2$) is more stable than 1S ($L = 0$).

$$^3P < ^1D < ^1S \quad (\text{increasing energy})$$

3.2 J Energy Levels for p^2

For 3P ($L = 1, S = 1$):

$$J = 2, 1, 0 \Rightarrow ^3P_2, ^3P_1, ^3P_0$$

For 1D ($L = 2, S = 0$):

$$J = 2 \Rightarrow ^1D_2$$

For 1S ($L = 0, S = 0$):

$$J = 0 \Rightarrow ^1S_0$$

Increasing order of energy:

$$^3P_0 < ^3P_1 < ^3P_2 < ^1D_2 < ^1S_0$$

Ground state Russell–Saunders term: 3P_0 .

4 Microstates

In a given configuration, electrons can be arranged in many different ways that have slightly different energy. These different arrangements are called **microstates**.

4.1 (1) Microstates from a Single Sub-shell

$$\text{Number of microstates} = \frac{n!}{r!(n-r)!}$$

where $n = 2 \times (\text{number of orbitals in subshell})$ and $r = \text{number of electrons}$.

Example – p^2 : $r = 2, n = 2 \times 3 = 6$

$$\frac{6!}{2!4!} = 15$$

Example – d¹: $r = 1$, $n = 2 \times 5 = 10$

$$\frac{10!}{1!9!} = 10$$

Table 3: Number of Microstates for p , d , and f Sub-shells

P^n Config	d^n Config	f^n Config
$P^1 = 6$	$d^1 = 10$	$f^1 = 14$
$P^2 = 15$	$d^2 = 45$	$f^2 = 91$
$P^3 = 20$	$d^3 = 120$	$f^3 = 364$
$P^4 = 15$	$d^4 = 210$	$f^4 = 1001$
$P^5 = 6$	$d^5 = 252$	$f^5 = 3003$
$P^6 = 1$	$d^6 = 210$	$f^6 = 3432$
	$d^7 = 120$	$f^7 = 3003$
	$d^8 = 45$	$f^8 = 3003$
	$d^9 = 10$	$f^9 = 2002$
	$d^{10} = 1$	$f^{10} = 1001$
		$f^{11} = 364$
		$f^{12} = 91$
		$f^{13} = 14$
		$f^{14} = 1$

4.2 (2) Microstates from Two Different Sub-shells

Multiply the microstates of each sub-shell.

Example – p¹d¹:

$$\text{For } p^1: n = 6, r = 1 \Rightarrow 6; \quad \text{For } d^1: n = 10, r = 1 \Rightarrow 10$$

$$\text{Total microstates} = 6 \times 10 = 60$$

4.3 (3) Microstates from Term Symbol (without J)

$$\text{Microstates} = (2S + 1)(2L + 1)$$

Example – ²D: $2S + 1 = 2$, $L = 2$

$$(2)(2 \times 2 + 1) = 2 \times 5 = 10$$

Microstates for p² configuration:

$$^3P: (3)(2 \times 1 + 1) = 3 \times 3 = 9$$

$$^1D: (1)(2 \times 2 + 1) = 1 \times 5 = 5$$

$$^1S: (1)(2 \times 0 + 1) = 1 \times 1 = 1$$

$$\text{Total} = 9 + 5 + 1 = 15 \checkmark$$

4.4 Pigeonhole Diagram for p^2

Total microstates of $p^2 = 15$, grouped as:

m_ℓ	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
-1			↓	↓	↑↓		↑	↑		↑	↑		↓	↓	
0		↓		↑		↑		↑	↑		↓	↓		↓	↑↓
+1	↑↓	↑	↑			↑	↑		↓	↓		↓	↓		
M_L	+2	+1	0	-1	-2	+1	0	-1	+1	0	-1	+1	0	-1	0
M_S	0	0	0	0	0	+1	+1	+1	0	0	0	-1	-1	-1	0
M_J	+2	+1	0	-1	-2	+2	+1	0	+1	0	-1	0	-1	-2	0
	(A)					(B)									(C)

- **Group A** ($M_L = +2, +1, 0, -1, -2$): $L = 2$ (D), $S = 0$, $J = 2 \Rightarrow$ five 1D_2
- **Group B** ($M_L = +1, 0, -1$, $M_S = +1$): $L = 1$ (P), $S = 1$, $J = 2, 1, 0 \Rightarrow ^3P_2, ^3P_1, ^3P_0$
- **Group C** ($M_L = 0$): $L = 0$ (S), $S = 0$, $J = 0 \Rightarrow ^1S_0$

4.5 Microstates for d^2 Configuration

$n = 10$, $r = 2$:

$$\frac{10!}{2!8!} = 45$$

Term symbols for d^2 : $^3F, ^3P, ^1G, ^1D, ^1S$

$$^3F: 3(2 \times 3 + 1) = 3 \times 7 = 21$$

$$^3P: 3(2 \times 1 + 1) = 3 \times 3 = 9$$

$$^1G: 1(2 \times 4 + 1) = 1 \times 9 = 9$$

$$^1D: 1(2 \times 2 + 1) = 1 \times 5 = 5$$

$$^1S: 1(2 \times 0 + 1) = 1 \times 1 = 1$$

$$\text{Total} = 21 + 9 + 9 + 5 + 1 = 45 \checkmark$$

Ground state term symbols for d^2 by Hund's rule:

$$^3F_2 < ^3F_3 < ^3F_4 < ^3P_0 < ^3P_1 < ^3P_2 < ^1G_4 < ^1D_2 < ^1S_0$$

4.6 (4) Microstates from Term Symbol with J

$$\text{Microstates} = (2J + 1)$$

Example – 2P_4 : $J = 4$

$$(2 \times 4 + 1) = 9$$

5 Electronic Spectra of Transition Metal Complexes

When electrons are promoted from one energy level to another, spectra arise. These are high-energy transitions. Much lower energy vibrational and rotational transitions always occur simultaneously, resulting in considerable broadening of electronic absorption bands. Band widths are commonly 1000–3000 cm^{-1} .

Not all theoretically possible electronic transitions are observed. Selection rules distinguish between ‘allowed’ and ‘forbidden’ transitions.

5.1 Selection Rules

5.1.1 (1) Spin Selection Rule

During a transition, the total spin of the molecule (atom or ion) must remain constant.

$$\Delta S = 0 \Rightarrow \text{allowed}; \quad \Delta S \neq 0 \Rightarrow \text{forbidden}$$

Allowed: singlet $\rightarrow$ singlet, doublet $\rightarrow$ doublet, triplet $\rightarrow$ triplet.

Forbidden: singlet $\rightarrow$ doublet, doublet $\rightarrow$ triplet, etc.

5.1.2 (2) Laporte Orbital Selection Rule

$$\Delta \ell = \pm 1 \Rightarrow \text{allowed (sharp band)}; \quad \Delta \ell \neq \pm 1 \Rightarrow \text{forbidden}$$

Allowed: s $\rightarrow$ p, p $\rightarrow$ s, p $\rightarrow$ d, d $\rightarrow$ p, d $\rightarrow$ f, f $\rightarrow$ d.

In terms of symmetry: $g \rightarrow u$ and $u \rightarrow g$ are allowed; $g \rightarrow g$ and $u \rightarrow u$ are forbidden.

Orbital	ℓ	Symmetry
s	0	gerade
p	1	ungerade
d	2	gerade
f	3	ungerade

5.2 Intensity of Electronic Transitions

Table 4: Approximate Molar Absorption Constants (ϵ)

Type of Transition	Approx. ϵ ($\text{L mol}^{-1} \text{ cm}^{-1}$)
Spin forbidden, Laporte forbidden	10^{-2} to 1.0
Spin allowed, Laporte forbidden	1 to 10
Spin allowed, Laporte forbidden (p & d orbital overlap)	10 to 10^2
Spin allowed, Laporte forbidden (intensity stealing)	10^2 to 10^3
Spin allowed, Laporte allowed	10^4 to 10^5

5.3 Splitting of d^n Terms

Each free ion term is affected by ligands in a complex. The effect depends on the geometry of the complex.

- s-orbitals: spherically symmetrical, unaffected.
- p-orbitals: all three affected equally, no splitting.
- d-orbitals: split into two states in octahedral/tetrahedral fields.
- f-orbitals: split into three states.

5.4 Transformation of Spectroscopic Terms to Mulliken Symbols

Spectroscopic Term	Octahedral Field	Tetrahedral Field
S	A_{2g}	A_1
P	T_{1g}	T_1
D	$E_g + T_{2g}$	$E + T_2$
F	$A_{2g} + T_{1g} + T_{2g}$	$A_2 + T_1 + T_2$

6 Electronic Spectra of Specific Systems

6.1 d^1 System

In the free gaseous metal ion, five d orbitals are degenerate and no $d-d$ transitions occur.

In octahedral complexes of Ti(III) (d^1 system) such as $[\text{TiCl}_6]^{3-}$ and $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, the d orbitals split and the one electron occupies the lower t_{2g} level. The only possible transition is to the e_g level.

Absorption spectrum of $[\text{Ti}(\text{H}_2\text{O})_6]^{2+}$ shows one band with peak at 20300 cm^{-1} .

Peaks for various Ti(III) complexes:

- $[\text{TiCl}_6]^{3-}$: 13000 cm^{-1}
- $[\text{TiF}_6]^{3-}$: 18900 cm^{-1}
- $[\text{Ti}(\text{H}_2\text{O})_6]^{2+}$: 20300 cm^{-1}
- $[\text{Ti}(\text{CN})_6]^{3-}$: 22300 cm^{-1}

6.2 d^9 System

Cu^{2+} ($Z = 29$): $[\text{Ar}] 3d^9 4s^0$, term symbol 2D .

In $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, the octahedral field splits d orbitals into lower $^2t_{2g}$ (d_{xy}, d_{yz}, d_{xz}) and higher 2e_g ($d_{x^2-y^2}, d_{z^2}$).

According to CFT, only one transition $^2T_{2g} \rightarrow ^2E_g$ should be observed (one band). However, the spectrum of $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ shows two broad, asymmetric bands. This is explained by the **Jahn–Teller effect**.

According to Jahn–Teller theory, a non-linear molecule containing degenerate orbitals in its ground state is distorted, destroying the degeneracy. Thus, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ loses its symmetry and the energy levels split further, giving three transitions:

1. $^2B_{1g} \rightarrow ^2E_g$

$$2. {}^2A_{1g} \leftarrow {}^2E_g$$

$$3. {}^2B_{2g} \leftarrow {}^2E_g$$

The ${}^2B_{2g} \leftarrow {}^2E_g$ transition is very low energy (near IR), so only two bands are observed in the visible region. These two bands appear close together as a broad peak with a shoulder. The result is that $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ appears blue-green in colour.

6.3 Correlation Diagrams

6.3.1 d^1 Correlation Diagram

Ground term: 2D . Ground state configuration: $t_{2g}^1 e_g^0$; excited state: $t_{2g}^0 e_g^1$. Transition: ${}^2T_{2g} \rightarrow {}^2E_g$, with energy equal to Δ_o .

Relative to the 2D free ion:

$${}^2T_{2g} \text{ stabilised by } -0.4\Delta_o; \quad {}^2E_g \text{ destabilised by } +0.6\Delta_o$$

6.3.2 d^9 Correlation Diagram

Ground term also 2D . Ground state: $t_{2g}^6 e_g^3$ (hole in e_g). Excited state: $t_{2g}^5 e_g^4$ (hole in t_{2g}). The energy ordering is reversed relative to d^1 :

$${}^2E_g \text{ stabilised by } -0.4\Delta_o; \quad {}^2T_{2g} \text{ destabilised by } +0.6\Delta_o$$

6.4 Orgel Diagram of $d^1 - d^9$

Both d^1 and d^9 have ground state term 2D . In d^1 the electron transitions; in d^9 the positron (hole) transitions. Absorption bands of both d^1 and d^9 occur at equal frequencies.

In a tetrahedral field, the energy level diagram for d^1 complexes is the inverse of that in an octahedral field and similar to d^9 .

6.5 d^2 Correlation Diagram

V(III) octahedral complexes have the d^2 configuration. Ground state term: 3F . Excited states: 3P , 1G , 1D , 1S .

By the spin selection rule, ${}^3F \rightarrow {}^3P$ is allowed; 1G , 1D , 1S are forbidden.

Three absorption bands arise from:

1. ${}^3T_{1g}(F) \rightarrow {}^3T_{2g}(F)$: 17200 cm^{-1}
2. ${}^3T_{1g}(F) \rightarrow {}^3T_{1g}(P)$: } 25600 cm^{-1}
3. ${}^3T_{1g}(F) \rightarrow {}^3A_{2g}(F)$: }

Because the energy difference between transitions (ii) and (iii) is very small, they overlap to give a single peak. Only two peaks appear in the spectrum of $[\text{V}(\text{H}_2\text{O})_6]^{3+}$.

6.6 d^8 Correlation Diagram

Ni(II) octahedral complexes have the d^8 configuration. Ground state term: 3F . Excited states: 3P , 1G , 1D , 1S .

In octahedral field, the 3F term splits into lower energy $^3A_{2g}$ and higher energy $^3T_{2g}$ and $^3T_{1g}(P)$ levels.

Three transitions in $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$:

1. $^3A_{2g} \rightarrow ^3T_{2g}(F)$: 8700 cm^{-1}
2. $^3A_{2g} \rightarrow ^3T_{1g}(F)$: 14500 cm^{-1}
3. $^3A_{2g} \rightarrow ^3T_{1g}(P)$: 25300 cm^{-1}

All three transitions are energetically sufficient, so the visible spectrum of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ contains three peaks.

6.7 d^2 – d^8 Orgel Diagram

Both d^2 and d^8 have ground state term 3F and excited states 3P , 1D , 1S .

The splitting of d^2 in octahedral field equals that of d^8 in tetrahedral field. The splitting of d^2 in tetrahedral field equals that of d^8 in octahedral field.

6.8 Electronic Spectra of High-Spin d^5 Ions

For high-spin d^5 ions, all possible d – d transitions are spin-forbidden, since any transition must involve a reversal of spin. As a result, bands in the spectra of high-spin Mn(III) and Fe(III) complexes are very weak and the compounds are nearly colourless.

The four quartets 4G , 4F , 4D , 4P involve the reversal of only one spin; the other seven states are doublets and doubly spin-forbidden. Up to ten extremely weak absorption bands may be observed in an octahedral field.

6.9 d^3 – d^7 Orgel Diagram

Cr(III) octahedral complexes have d^3 ; Co(III) octahedral complexes have d^7 . Ground state term: 4F ; excited states: 4P , 2G , 2F , 2D , 2P .

By spin multiplicity rule, $^4F \rightarrow ^4P$ is allowed; 2G , 2F , 2D , 2P are forbidden.

$[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ shows three absorption bands:

1. $^4T_{1g}(F) \rightarrow ^4T_{2g}(F)$: 8000 cm^{-1}
2. $^4T_{1g}(F) \rightarrow ^4A_{2g}(F)$: 19600 cm^{-1}
3. $^4T_{1g}(F) \rightarrow ^4T_{1g}(P)$: 21600 cm^{-1}

Tetrahedral $[\text{CoCl}_4]^{2-}$ shows intense blue colour. Possible transitions:

1. $^4A_2(F) \rightarrow ^4T_1(P)$: 15000 cm^{-1}
2. $^4A_2(F) \rightarrow ^4T_1(F)$: 5800 cm^{-1}
3. $^4A_2(F) \rightarrow ^4T_2(F)$: 3500 cm^{-1}

Dissolving CoCl_2 in water produces a pale pink solution of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, whereas in alcohol the tetrahedral $[\text{CoCl}_2(\text{CH}_3\text{CH}_2\text{OH})_2]$ forms with intense blue colour.

The tetrahedral complex $[\text{CoCl}_4]^{2-}$ shows much more intense $d-d$ bands than the octahedral $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ because the tetrahedral complex has no centre of symmetry, helping to overcome the $g \rightarrow g$ Laporte selection rule.

6.10 Orgel Diagram of d^4-d^6

Ground state term for both d^4 and d^6 is 5D . Excited states: 3h , 3G , 3F , 3D , 3P , 1I , 1D , 1S .

Only the 5D transition is spin-allowed. The 5D term splits into T_{2g} and E_g .

- High-spin d^6 : electron promoted $T_{2g} \rightarrow E_g$ (same as d^1).
- d^4 : positron (hole) promoted $E_g \rightarrow T_{2g}$ (same as d^9).

In a tetrahedral field, the subscript g is omitted from the Orgel diagram.

The combined Orgel diagram for d^1 , d^9 , d^4 , d^6 systems shows analogous splitting patterns.

7 Jahn–Teller Effect

The Jahn–Teller theorem explains why certain six-coordinate complexes possess distorted octahedral geometry.

1. An octahedral complex has a regular shape when both t_{2g} and e_g sets of d -orbitals are symmetrically occupied.
2. When the t_{2g} orbitals are asymmetrically occupied, **slight distortion** occurs. This happens when the central metal ion contains 1, 2, 4, or 5 electrons in t_{2g} .
3. When the e_g orbitals are asymmetrically filled, **strong distortion** occurs, potentially changing the octahedral shape to tetragonal or even square planar. High-spin d^4 and d^9 , and low-spin d^7 , d^8 , d^9 configurations lead to strong distortion.

7.1 Explanation for Cu^{2+} (d^9)

Configuration: $t_{2g}^6 e_g^3$. Two possible arrangements:

1. $t_{2g}^6 (d_{z^2})^2 (d_{x^2-y^2})^1$
2. $t_{2g}^6 (d_{z^2})^1 (d_{x^2-y^2})^2$

Due to unsymmetrical e_g occupation, ligands along the z -axis move away from the nucleus, while ligands in the x - y plane move closer. Thus, Cu^{2+} complexes have **four short and two long** metal–ligand bonds.

Examples:

- Cupric chloride crystal: four Cl^- at 2.30 Å, two Cl^- at 2.95 Å.
- Cupric fluoride crystal: four F^- at 1.93 Å, two F^- at 2.27 Å.

In the Jahn–Teller splitting: $\Delta_o \gg \delta_1 > \delta_2$.

8 Spectrochemical Series

The relative strength of different ligands can be determined by examining the electronic spectra of their complexes. When a stronger ligand replaces a weaker one, the absorption band shifts to **shorter wavelength** (blue shift, higher energy). A weaker ligand causes a **red shift** (longer wavelength, lower energy).

The spectrochemical series, in order of increasing ligand strength, is:



9 Assignment

Short Questions

1. Define term symbol.
2. Give the value of L for s , p , d , f , and g .
3. What is spin multiplicity?
4. Give Pauli's Exclusion Principle and Hund's Rule.
5. Define: Total degeneracy.
6. Write the rules for the ground state term symbol.
7. Find the ground state term symbol for ${}^7\text{N}$ ($Z = 7$), ${}^6\text{C}$ ($Z = 6$), V^{3+} ($Z = 23$), Mn^{2+} ($Z = 25$), Ni^{2+} ($Z = 28$), Cu^{2+} ($Z = 29$).
8. Find L , S , spin multiplicity, number of unpaired electrons and J values of 3F , 1S , 5D .
9. Calculate microstates for p^3 and d^2 .
10. Explain: Bands of $d-d$ transition are weaker.
11. Explain: Identification of octahedral and tetrahedral complexes through electronic spectra.
12. According to the Laporte rule, transition metal complexes should be colourless, but this is not true. Explain.
13. Complexes of Zn^{2+} are mostly colourless. Explain.
14. Explain: The electronic absorption band of *cis*- MA_4B_2 is more intense than *trans*- MA_4B_2 .

Long Questions

1. Explain the R-S (L-S) coupling scheme to derive the ground state term symbol.
2. What is L-S coupling? Explain with the example of the p^2 configuration of carbon.
3. Derive the ground state term symbol for: (i) ${}^7\text{N}$, (ii) ${}^{16}\text{S}$, (iii) ${}^{22}\text{Ti}$, (iv) Cr^{3+} ($Z = 24$), (v) Fe^{3+} ($Z = 26$), (vi) Cu^{2+} ($Z = 29$), (vii) Co^{2+} ($Z = 27$), (viii) Ni^{2+} ($Z = 28$).
4. Identify the ground state term (giving reason) for: (i) 1S , (ii) 3F , (iii) 3P , (iv) 1G , (v) 1D .
5. Calculate L for p^2 configuration and show the energy states.
6. Draw the pigeonhole diagram for p^2 configuration and arrange terms according to energy level.
7. Draw the pigeonhole diagram for d^2 configuration and arrange terms according to energy level.
8. Discuss the Selection Rules for electronic transition.
9. Explain the Jahn-Teller effect in $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$.
10. Discuss the electronic spectra of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ complex ion.
11. Explain the Jahn-Teller effect in $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$.
12. Explain: Orgel energy diagrams of d^1 and d^9 are opposite to each other.
13. Write a short note on hole formalism.
14. In $[\text{V}(\text{H}_2\text{O})_6]^{3+}$, three transitions are possible but only two peaks are observed. Explain.
15. In $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, three peaks are observed. Explain.
16. Draw and explain the combined Orgel diagram of d^3 and d^7 .